FoldPulse

Every Fold. Every Beat.

PHASE 2 · BRAND, UI/UX & DESIGN

Design Report & Specifications

Version	1.0
Date	9 November 2026
Prepared for	Founder & Product Owner
Prepared by	FoldPulse product team (12 agent roles)
Status	Submitted for Founder approval
Recommendation	**APPROVE** (open items R-28, I-24 recorded)

Confidential — prepared for the FoldPulse Founder. All numbers in mockups are sample data.

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1 Executive summary & recommendation

PHASE 2 RECOMMENDATION

APPROVE

The design is complete and reviewed. Two open items are recorded (R-28, I-24); neither blocks Phase 3.

What this phase delivers

FoldPulse now has a complete, reviewed design, ready to be built:

- **Brand and final app icon:** an open foldable crossed by a pulse line, with default, dark and tinted variants, tested down to 40 px.
- **A design system:** colour tokens with **measured** contrast (all 36 text pairs pass WCAG AA), type, spacing, components, motion and accessibility rules.
- **24 high-fidelity mockups** in light and dark, for the outer (5.4") and inner (7.6") displays: onboarding, Today (including the honest "background counting not set up" state), Record, History, Edit, Insights, Coverage, background-counting setup, Settings, Export, Milestone and manual-only mode.
- **Widgets:** Home Screen small (also used in StandBy) and Lock Screen circular and rectangular, with freshness, midnight and privacy rules.
- **User flows** for first launch, optional background setup with self-test, recording and correcting, export and reset, milestones and non-hinge devices.

Decisions this design follows

Decision	Effect on the design
D-5 Paid up front	No paywalls, upgrade prompts, tip jars or ads anywhere
D-6 Lean v1	Three widgets; no import, gap toggle, heat strip or session stats (tagged v1.1)
C-2r Device test skipped	Background counting is **optional and unverified**. The default "counting only while open" state is designed honestly, and setup reports status only through a self-test
D-2 Gaps excluded	Gaps are shown in History and Coverage, never added to totals

What changed after the agent reviews

The independent reviewers found real problems, and all of them were fixed before this report:

- **Setup screen:** it implied FoldPulse could see the user's Shortcuts steps. It now shows plain numbered steps, and only the self-test reports a status.
- **Coverage metric:** it overclaimed. It is now "Counted for you · % of recorded folds", with no "Good" verdict.

- **Sample data:** it was inconsistent (wrong weekdays, totals that didn't add up, a misplaced gap row). It was regenerated from one seed.
- **Icon:** it read as an open book at small sizes. It was redesigned.
- **Fold rule and controls:** the fold-region rule was corrected, and custom controls were replaced by system controls.

Recommendation

Approve Phase 2. Two open items are recorded and not hidden:

1. **R-28 (High):** FoldPulse is a paid app whose background counting is unverified. The Founder Representative strongly recommends updating your iPhone Duo to **iOS 27.2** and running the 10-minute automation check (S-2/S-3) **before a price is set**.
2. **I-24:** UI Developer and Compliance disagree about the inner-display tab bar when the phone is partially folded. It will be checked in the Xcode simulator in Phase 3.

2 Inputs, team & method

Inputs

Phase 1 report and documents; decisions A-1, C-1, C-2r, D-1…D-6, I-17 (`DECISIONS.md`); partial device results (`SPIKE_RESULTS.md`): FoldProbe built unchanged, hinge callbacks fire continuously, and an informal 5/5 fold count was correct.

Who did what

Agent	Role in Phase 2	Mode
4 — UI/UX Design Director	Brand, icon, design system, screens, flows, widgets	Main session (author)
5 — Senior iOS UI Developer	Buildability with system SwiftUI components	**Independent subagent**
10 — Compliance & Accessibility	HIG, iPhone Duo guidance, widgets, App Review, accessibility	**Independent subagent** (with Agent 5)
2 — Founder Representative	Value, simplicity, paid-app expectations	**Independent subagent**
12 — Independent Final Reviewer	Consistency, overclaims, PRD alignment	**Independent subagent** (with Agent 2)
7 — Data Analyst	Sample-data consistency, coverage definition	Main session (applied fixes)
1 — Chief Product Manager	Scope alignment (PRD, MVP), issue log	Main session

How the mockups were made

The mockups are HTML/CSS built from the design-system tokens and rendered by headless Chromium at 2× (`design/mockups/`). Inter stands in for SF Pro. **Display proportions are assumptions** (outer 376 × 820 pt, inner 760 × 860 pt) to be checked in Xcode 27.1's iPhone Duo simulator in Phase 3. Bars, sheets and Liquid Glass are approximations of system components, which draw themselves in the real app.

Sample data (illustrative, never shown to users): installed **Fri Oct 24**, today **Mon Nov 9, 14:04**, 1,284 folds since install, last 7 days 58 · 77 · 69 · 81 · 74 · 73 · 64 (today so far), full-day average 72, busiest day Fri Nov 6 (81), coverage this week 303 automatic + 164 shortcut + 29 manual = 496 folds, 2 gaps.

3 Brand & app icon



Figure 1 — Final icon: default, dark and tinted appearances; small-size test; Home Screen

Why this icon: an open foldable (two panels, hinge seam) crossed by a single pulse line, the "beat" of each fold. It contains no text and no Apple device silhouette or trademark. It was redesigned after the review: the Phase 1 V-shaped panels read as an open book at 40 px. The final layered version (background · panels · pulse) is assembled in **Icon Composer** in Phase 3, so the system can apply Liquid Glass highlights and the dark and tinted appearances.

Brand voice: plain, factual, warm. Counts are always "since install". No wear or health language, no gamification. Tagline: **Every Fold. Every Beat.**

4 Design system

1. Brand

Element	Decision
Name	**FoldPulse** (one word, capital F and P)
Tagline	**Every Fold. Every Beat.**
Personality	A precise, calm instrument. Native, premium, quiet. Closer to Weather or Fitness than to a game
Voice	Plain, factual, warm. Numbers first. Never alarmist about the hinge, never gamified
Signature motif	The **pulse line**: a short ECG-like stroke that "beats" once when a fold is recorded
Business model impact	Paid up front (D-5): **no paywalls, upsells, tip prompts or ads** anywhere in the UI

1.1 Copy rules

- Counts are always **"since install"** or **"since tracking began"**. Never "lifetime of your phone".
- No wear or health language ("hinge health", "wear", "lifespan", "damage"). FoldPulse counts; it does not diagnose.
- Every number can say **where it came from** (Automatic / Shortcut / Manual / Gap).
- Background counting is described as **optional**: "Count in the background with a Shortcuts automation".
- Stale data says so: "as of 14:02".
- Numbers use locale grouping (1,284 / 1.284) and monospaced digits.

2. App icon (final recommendation)

The final icon refines **Concept A, "Fold + Pulse"**: an **open foldable seen from the front**, with a left panel in Pulse Blue and a right panel in Signal Cyan, separated by a hinge seam, crossed by one white pulse line, on a deep-navy squircle. *Revision:* the Phase 1 V-shaped panels read as an open book at 40 px (Compliance review), so the panels are now upright and rounded like a phone. Files: `design/icon/` (SVG + 1024 px PNG for each appearance).

Variant	Use
Default (navy background)	Home Screen, App Store
Dark	iOS dark icon appearance: black background, brighter panels
Tinted	iOS tinted appearance: monochrome glyph (panels + pulse) on the system tint
Layer structure (Liquid Glass)	Built in **Icon Composer** with three layers: background, panels, pulse. The system adds specular highlights and depth

Category and keywords: Utilities. No health or medical keywords anywhere in the listing, because the pulse motif plus the name could otherwise be read as a health app (App Review 1.4.1 / 5.1.3 risk, R-27).

Small-size rule: at 29 pt the pulse line must stay ≥ 2 px thick and the panels must stay distinguishable. This was tested at 1024, 180, 120, 87, 58 and 40 px (Figure 2 of the Phase 2 report). The icon contains no text and no Apple device silhouette or trademarked shapes.

3. Colour

All colours are defined as **semantic tokens** in the asset catalog, with Light and Dark variants. Contrast was **measured** with `design/contrast_check.py` (WCAG 2.x). **All 36 text/background pairs pass AA**; the full table is in `design/contrast_report.md`.

Token	Light	Dark	Use	Lowest measured contrast vs `bg` / `surface` (light / dark)
`bg`	#F2F4F8	#070F1F	Screen background	—
`surface`	#FFFFFF	#12233F	Cards, tiles	—
`text`	#0B1B33	#FFFFFF	Primary text, hero numbers	15.65:1 / 15.70:1
`text2`	#55606E	#9FB1CF	Secondary labels	5.81:1 / 7.22:1
`accent`	#2F7BFF	#4C8DFF	Pulse Blue: charts, large numerals, icons, buttons (fills)	3.53:1 / 4.90:1 (graphics & large text only; needs 3:1)
`accentText`	#1D5FD8	#6FA2FF	Accent-coloured **small** text and links	5.18:1 / 6.19:1
`src.automatic`	= accent	= accent	Automatic (hinge) events	—
`src.shortcut`	#0B6F85	#5CC8FF	Shortcut-automation events	5.26:1 / 8.34:1
`src.manual`	#55606E	#B4C0D3	Manual events	5.81:1 / 8.54:1
`src.gap`	outline only, `text2`	outline only	Inferred gaps (dashed)	—
`positive`	#1E7F4F	#5BD08E	Coverage good (≥ 90%)	4.54:1 / 8.12:1
`warning`	#A15C00	#FFB547	Coverage low, setup needed	4.71:1 / 8.93:1
`destructive`	#C2362B	#FF6B5E	Delete, reset	4.95:1 / 5.62:1

Rule: colour is never the only carrier of meaning. Source badges pair colour with an SF Symbol and a text label, and coverage pairs colour with a percentage and a word ("Good", "Low").

Gradient: Pulse Blue → Signal Cyan (#5CC8FF), used only for the hero progress bar and the icon.

4. Typography

The app uses **SF Pro** (the system font). All styles use **Dynamic Type** text styles, so they scale up to AX5. Mockups use Inter as a stand-in because SF Pro cannot be used outside Apple platforms.

Role	SwiftUI	Notes
Hero numeral	`.system(size: 64, weight: .bold, design: .rounded)` scaled by `@ScaledMetric(relativeTo: .largeTitle)`, `.monospacedDigit()`	Total since install; `.contentTransition(.numericText())`
Tile numeral	`.system(.title, design: .rounded).weight(.bold).monospacedDigit()`	Today, averages
Screen title	`.largeTitle` (system navigation title)	
Section header	`.headline`	
Body	`.body`	
Label / meta	`.subheadline`, `.footnote` in `text2`	
Badge	`.caption.weight(.semibold)`	

At accessibility sizes (≥ AX1), tiles stack vertically. The hero numeral uses `ViewThatFits` with a ceiling on the scaled size and `minimumScaleFactor(0.5)` as a last resort: a 6-digit number cannot wrap on the 5.4" display. Text in rows is never truncated; rows grow in height instead.

5. Layout, spacing and shape

- **Grid:** 4-pt base. Spacing scale: 4 · 8 · 12 · 16 · 20 · 24 · 32.
- **Margins:** system readable margins (`.scenePadding()`); cards use 16 pt inner padding.
- **Corners:** cards 20 pt, tiles 16 pt, badges 8 pt, all `RoundedRectangle(style: .continuous)`.
- **Depth:** no drop shadows. Hierarchy comes from surface colour, system **materials** and **Liquid Glass** on system bars.
- **Displays:** layouts are built from **size classes and container size**, never device checks (Apple: *Preparing your app for iPhone Duo*).
  - **Outer display** (compact width): one column. The system may present the tab bar **vertically along the side**, so FoldPulse uses `TabView` and never draws custom bars.
  - **Inner display** (regular width): two panes built with **`ArrangementView` (split style)**, with primary content and a secondary pane. When **fully open**, the screen has no fold division, so content may span it; the split is a layout choice, not a hinge workaround.
  - **Partially folded ("tent" / "laptop")**: the fold becomes an active reserved region (`.divisions`). `ArrangementView` moves the panes onto either side, and controls avoid the fold. *Corrected after the UI Developer review: the fold is reserved only when partially open.*

6. Components

Component	Description	States
`HeroCounter`	"Folds since install" label + hero numeral + pulse line	normal, counting animation, empty ("0" + setup prompt)
`StatTile`	Small card: label, numeral, optional caption	normal, no-data ("—")

Component	Description	States
`CoverageTile`	**"Counted for you"**: % **of recorded folds** captured without tapping + stacked bar by source; tap → coverage detail. *Renamed after review:* "captured automatically" clashed with the "Automatic" source name and overclaimed (missed folds can't be in the denominator)	high (≥ 90%, `positive`) · medium (70–89%, `text2`) · low (< 70%, `warning`) · not set up (`warning` card)
`MilestoneBar`	Gradient progress bar toward the next milestone, with "N to go"	normal, reached
`SourceBadge`	Symbol + label: Automatic · Shortcut · Manual · Gap	4 variants; also icon-only for compact rows
`EventRow`	Time, type (Fold/Unfold), source badge, note indicator	normal, edited, gap (dashed)
`WeekChart`	Swift Charts bar chart, 7 days; today highlighted	data, partial, empty
`RecordButton`	System `Button` with `.buttonStyle(.glassProminent)` (fallback `.borderedProminent`) and `.controlSize(.extraLarge)`; not hand-drawn	enabled, just-recorded (Undo)
`FreshnessLabel`	"Updated just now" / "as of 14:02"	live, stale
`SetupCard`	Inline call to action for background counting	not set up, verified, failed self-test
`EmptyState`	`ContentUnavailableView` with action	—
Filters	Source filter = segmented `Picker`; date range = toolbar `Menu` + `DatePicker`. No custom chips	—
Lists	`List` + `.swipeActions` (VoiceOver actions come for free)	—

7. Iconography

The app uses **SF Symbols** only, so they inherit weight and Dynamic Type. Proposed symbols (names to be confirmed in the SF Symbols app for iOS 27):

Concept	Symbol
Today tab	`waveform.path.ecg`
History tab	`list.bullet.rectangle`
Insights tab	`chart.bar.xaxis`
Settings tab	`gearshape`
Automatic	`sensor.fill`
Shortcut	`square.2.layers.3d`
Manual	`hand.tap`
Gap	`questionmark.circle.dashed` (fallback `questionmark.circle`)
Fold / Unfold	`arrow.down.right.and.arrow.up.left` / `arrow.up.left.and.arrow.down.right`

8. Motion and haptics

Moment	Motion	Haptic	Reduce Motion
Automatic fold recorded (app open)	Hero numeral `.numericText()` roll + one pulse-line beat (600 ms)	**None**: avoids noise across ~100 folds a day	Crossfade only
Manual record	Same + button ripple; Undo toast. **Undo stays until dismissed when VoiceOver, Switch Control or Voice Control is running** (no timing barrier, WCAG 2.2.1), and an announcement is posted. Undo is also always possible from History	`.sensoryFeedback(.increase)`	Crossfade, haptic kept
Milestone reached	Sheet with the pulse line drawing itself (1.2 s), once	`.success`	Static illustration
Delete event	System list animation	`.impact(weight: .light)`	System default

9. Accessibility checklist (applies to every screen)

- Dynamic Type through AX5 without truncation. Layouts reflow at AX sizes.
- VoiceOver: counters read as full sentences ("1,284 folds since install"). Charts expose **Audio Graphs** via `AXChartDescriptor`. Source badges read their label, including icon-only badges on the outer display (`accessibilityLabel`). The pulse animation is hidden from accessibility.
- Minimum hit target 44 × 44 pt. The Record buttons are 56 pt tall.
- Contrast AA, measured (§3). Supports Increase Contrast (higher-contrast token variants) and Smart Invert (images excluded).
- Reduce Motion and Reduce Transparency respected.
- No information conveyed by colour alone, sound alone or haptics alone.
- Every control is reachable with Full Keyboard Access and Voice Control, with visible labels.

5 Navigation & user flows

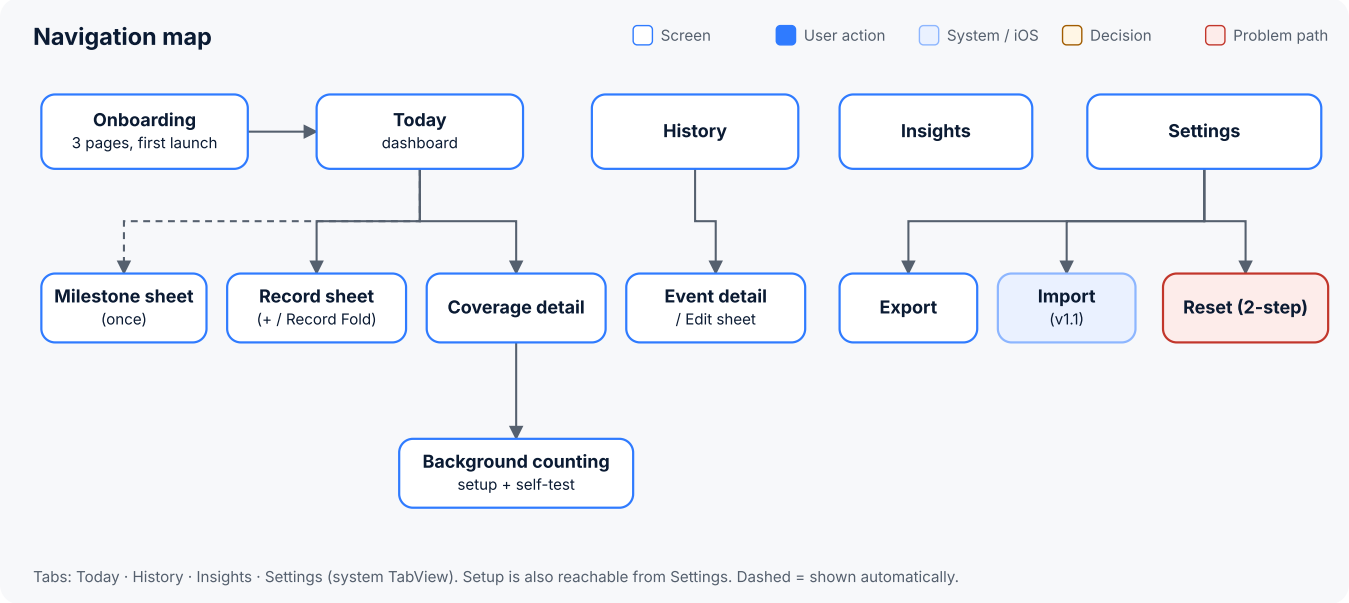


Figure 2 — Navigation map

F0. Navigation map (`flow-nav.svg`)

Four tabs: **Today**, **History**, **Insights**, **Settings**. Sheets: Record, Edit, Export, Milestone (Import arrives in v1.1). Full-screen cover: Onboarding (first launch only). Pushed screens: Coverage detail, Background counting setup, Event detail (outer display).

F1. First launch (`flow-setup.svg`, top row)

1. App opens → **Onboarding 1** (value) → **2** (how counting works: Automatic / Shortcut / Manual) → **3** (privacy) → **Start counting**.
2. **Today (empty state)**. Automatic counting is already active while the app is open. No permission prompts.
3. The user can fold right away and watch the count start ("Ready when you are").

Design intent: the user sees value within 30 seconds, without any setup. Background counting is offered but never forced.

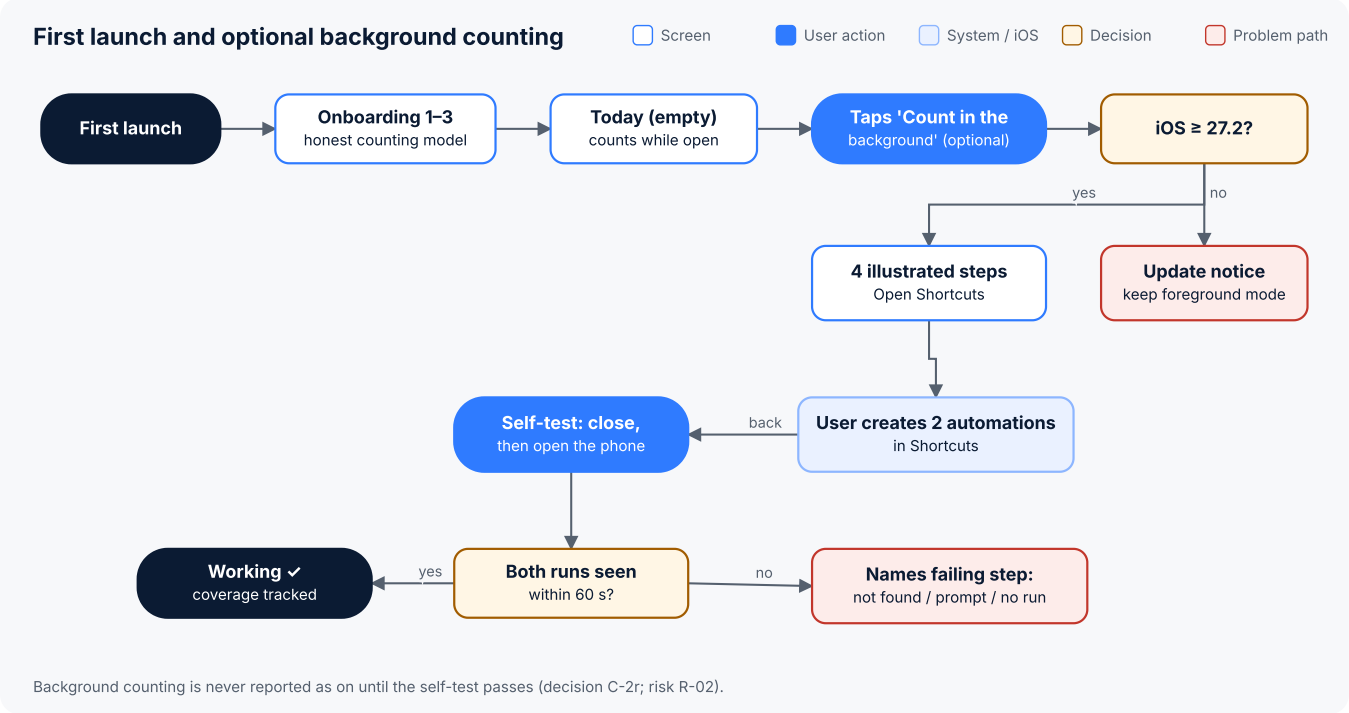


Figure 3 — First launch and optional background counting

F2. Optional background counting (`flow-setup.svg`)

1. Entry: the Today card "Count in the background", Coverage detail, or Settings.
2. **Version check:** on iOS < 27.2, show the update notice and keep foreground mode (the Phase 1b finding).
3. **Setup screen:** four plain numbered steps (FoldPulse can't observe Shortcuts, so no checkmarks) → **Open Shortcuts**.
4. In Shortcuts, the user creates **Closed → Record Fold** and **Opened → Record Unfold**, choosing **Run Immediately**.
5. Back in FoldPulse, the **self-test**: close the phone, then open it. FoldPulse looks for a shortcut fold and unfold within 60 s.
6. **Pass:** "Working ✓", and coverage starts tracking. **Fail:** the screen names the likely cause (trigger not found → iOS version; confirmation prompt → choose Run Immediately; no run → check the action is FoldPulse's) and offers to retry.

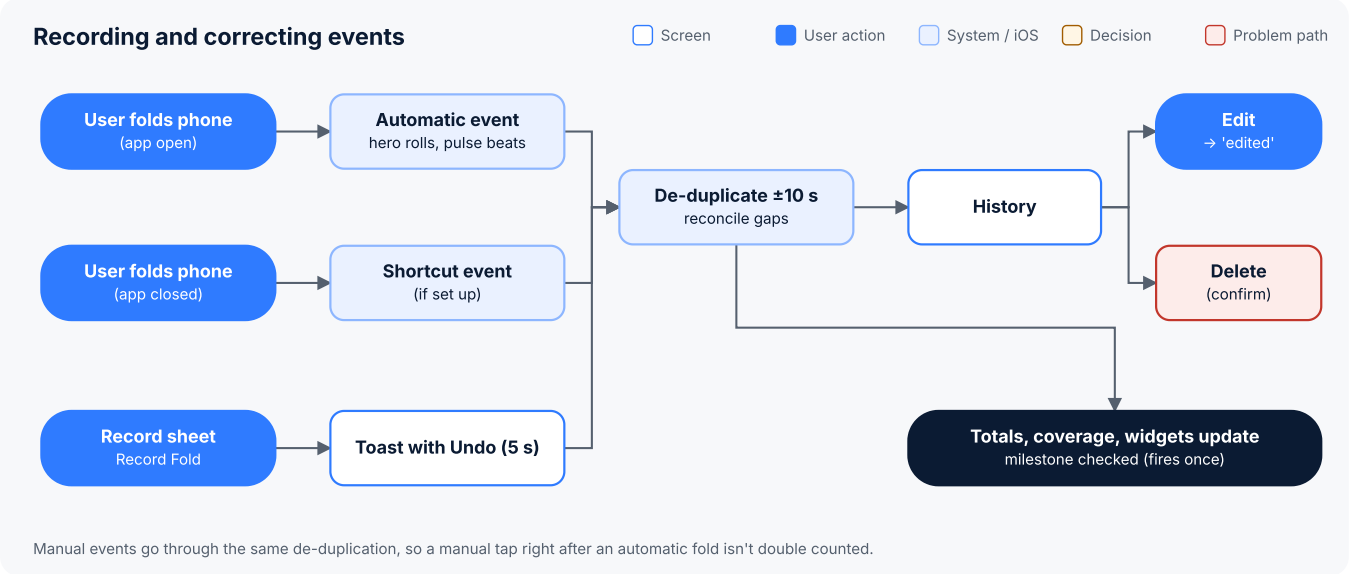


Figure 4 — Recording and correcting events

F3. Record and correct (`flow-record.svg`)

1. **Automatic** (app open): fold → event → hero rolls, pulse beats, no haptic.
2. **Shortcut** (app closed, if set up): the automation records in the background; widgets refresh within the budget.
3. **Manual:** Record sheet → Record Fold → toast with **Undo** (5 s).
4. All sources pass through **de-duplication (±10 s) and gap reconciliation** (feasibility §6).
5. **Correct:** History → tap → Edit (type, time, note; the event is marked "edited"), or swipe → Delete (confirmation for automatic and shortcut events).
6. Totals, coverage and widgets update, and the milestone check runs (each milestone fires once).

F4. Export and import

1. Settings → **Export data** → choose CSV/JSON and range → privacy note → share sheet → a temporary file is deleted afterwards.
2. *(v1.1, deferred by D-6)* Settings → **Import data** → file picker → validation (size, schema, bounds) → preview ("312 new, 14 already present, 0 invalid") → **Merge**. Nothing is written if validation fails.

F5. Reset

Settings → **Reset all data** → explanation ("This deletes all 2,565 events on this iPhone. Export first?") with an **Export first** shortcut → type **RESET** → done → first-run Today state. Widgets reload.

F6. Milestone

When the total crosses a milestone, the next time Today is visible a sheet appears **once**: pulse animation, number, date reached, next milestone. Optional local notification if enabled (off by default; permission asked only when the user turns it on).

F7. Non-hinge device

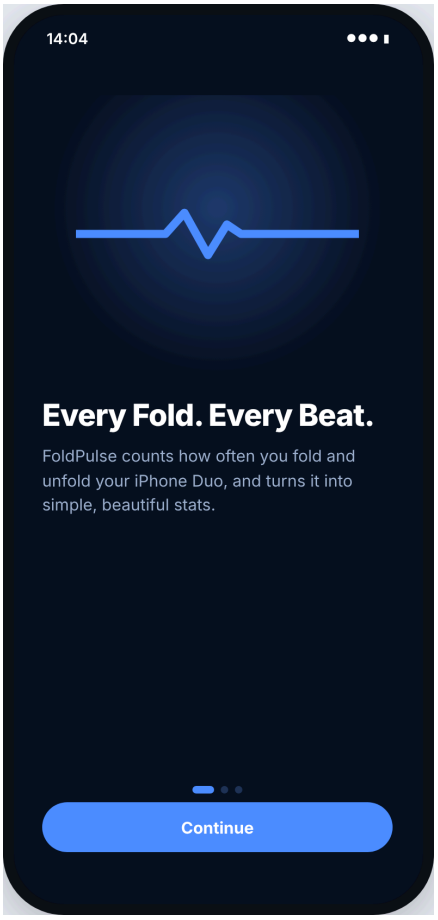
No hinge reading within 10 s on a model without a hinge → the manual-only explanation → manual mode. Coverage and background setup are hidden.

Flow-level acceptance checks (for QA, Phase 5)

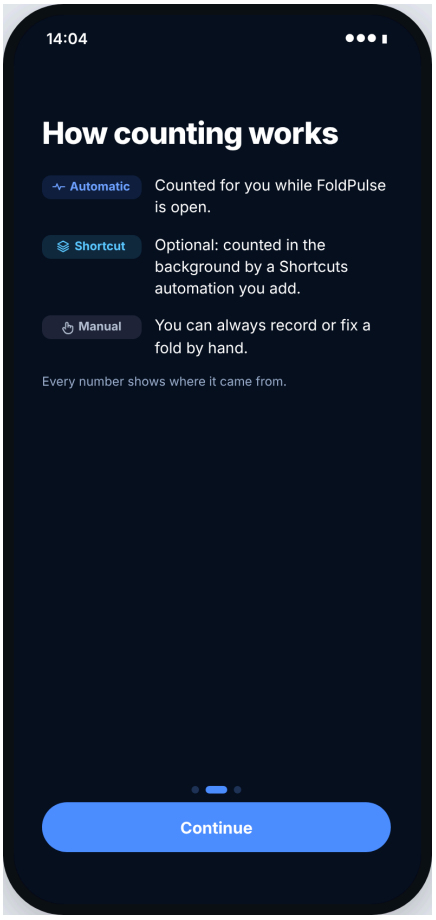
Flow	Check
F1	A new user can see their first automatic fold counted in under 30 s from launch
F2	The app never shows "Working" without a passed self-test; on iOS 27.1 it shows the update notice
F3	Folding with the Record sheet open, then tapping Record Fold within 10 s, leaves exactly **one** fold
F4	Export produces a CSV that opens safely in a spreadsheet (no formula execution). Import checks move to v1.1
F5	After reset, widgets show 0 within one reload
F6	Deleting events below a milestone never re-shows its celebration

6 Screen designs

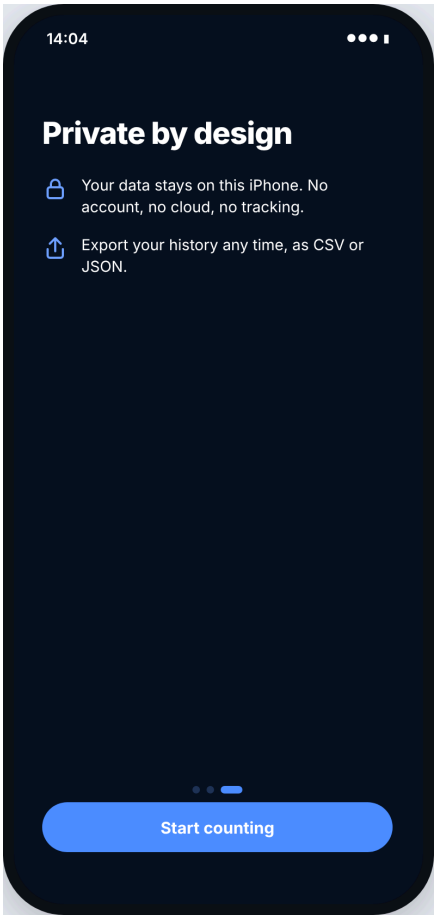
Sample data only. Outer display (5.4"), light and dark. The system presents bars vertically on the side of the outer display (Apple's iPhone Duo guidance); FoldPulse uses the standard `TabView` and toolbar items so this happens automatically.



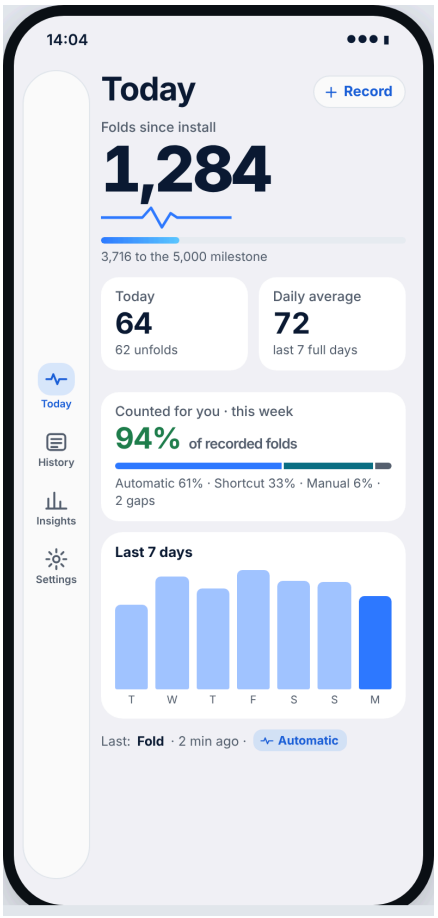
Onboarding 1 · welcome



Onboarding 2 · how counting works



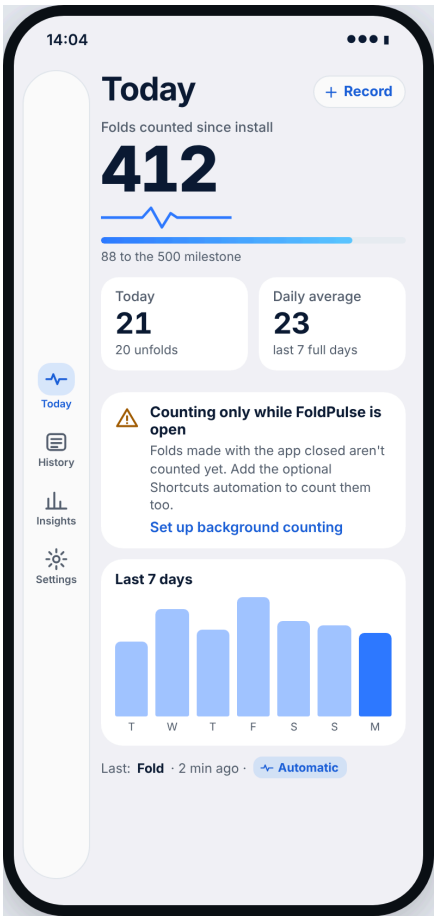
Onboarding 3 · privacy



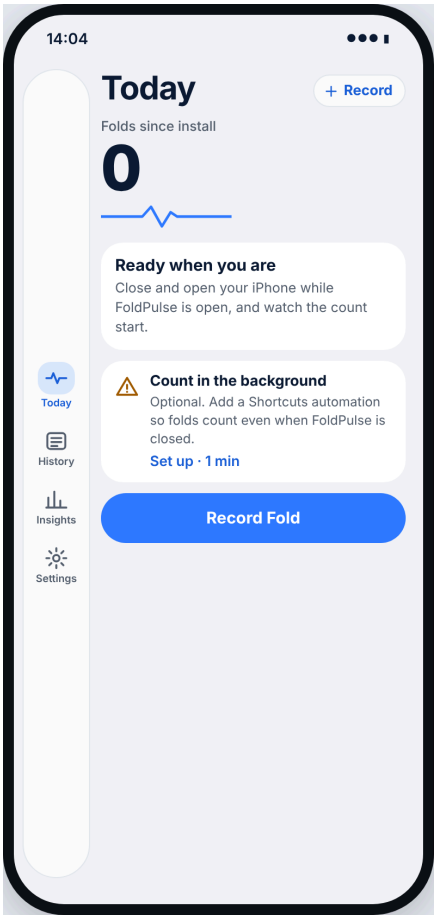
Today · light



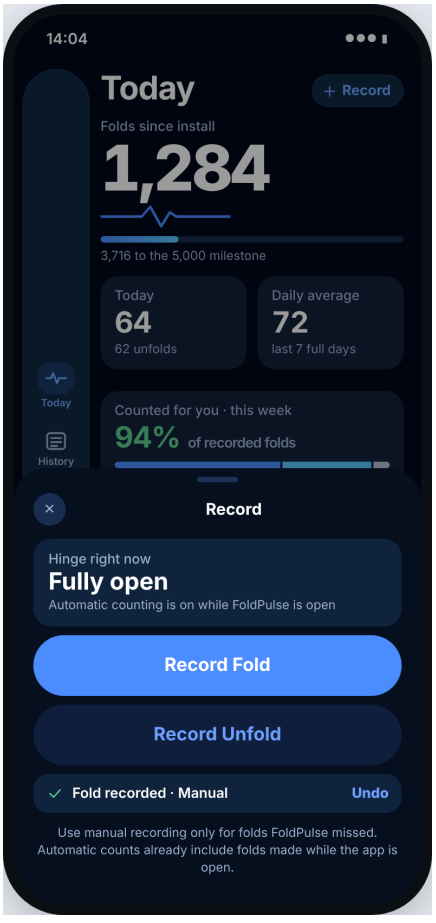
Today · dark



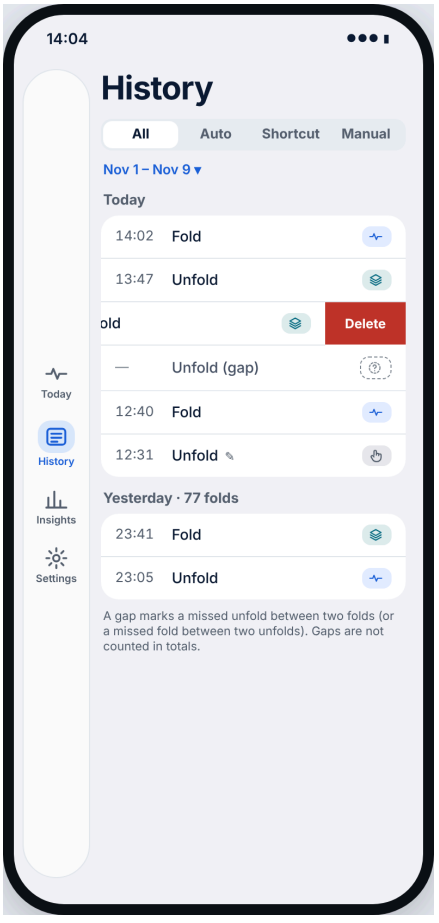
Today · background counting not set up



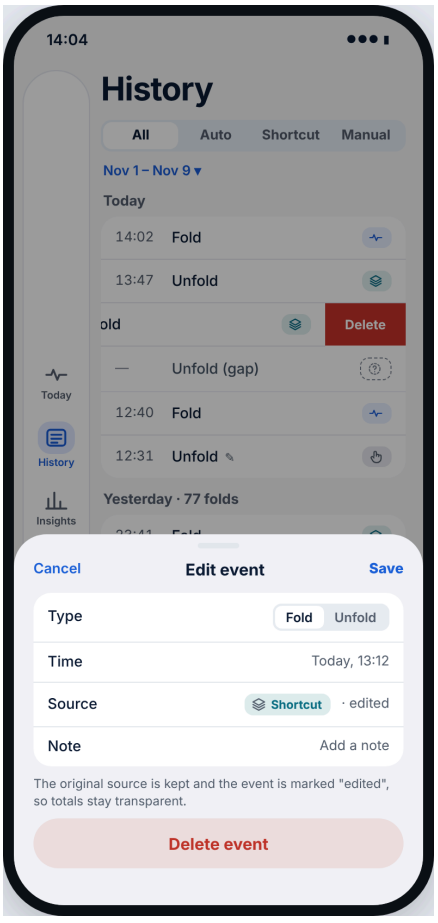
Today · first run



Record sheet + Undo



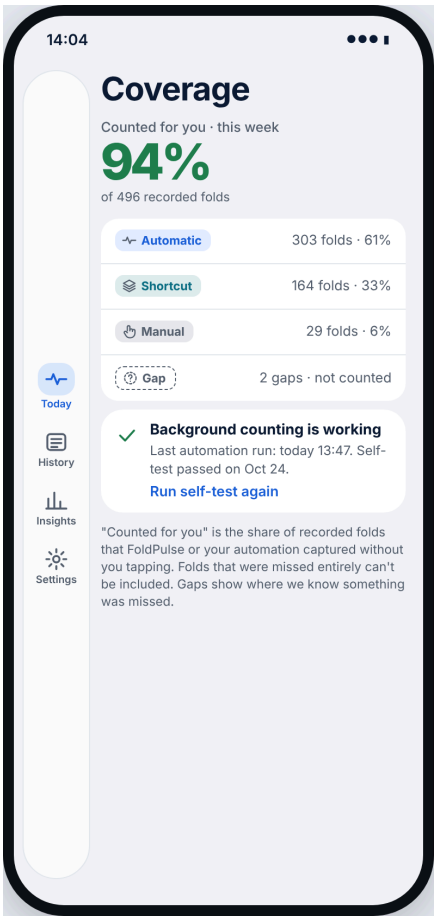
History · swipe to delete, gap row



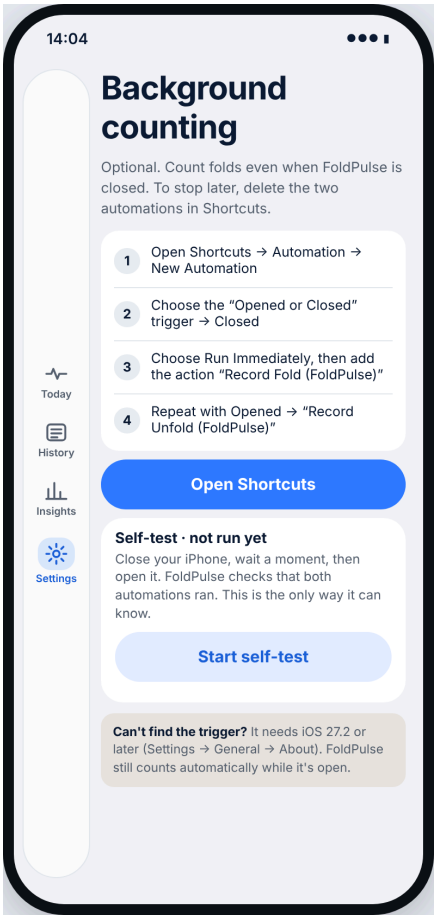
Edit event



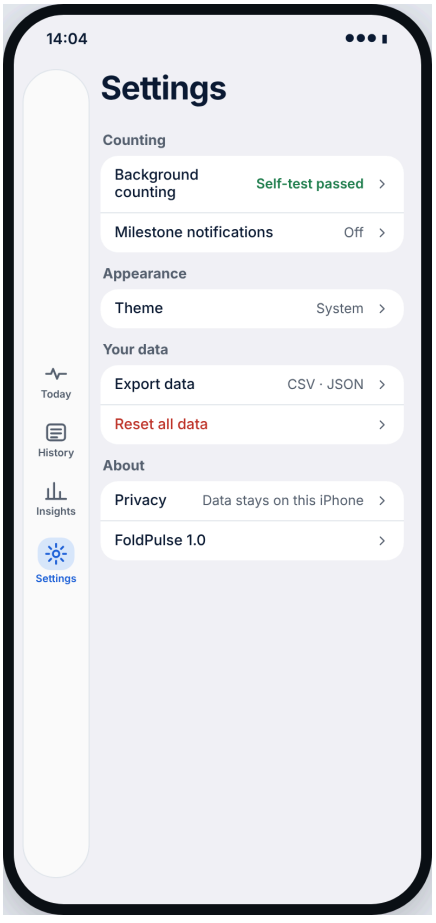
Insights · last 7 days



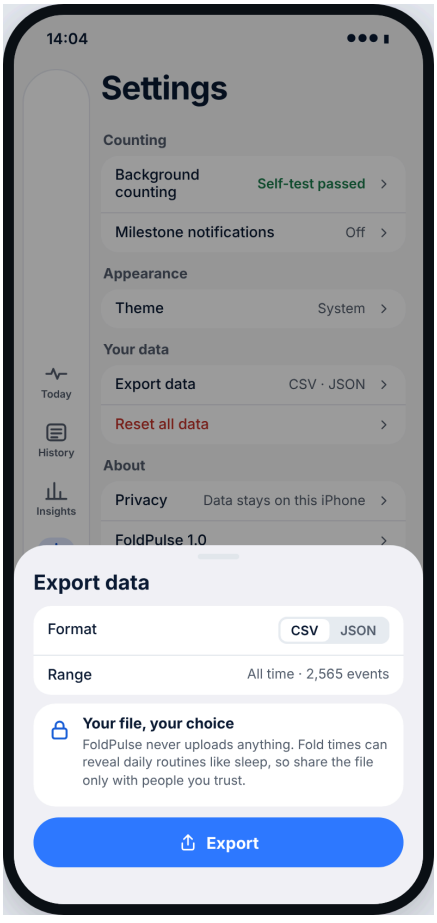
Coverage detail



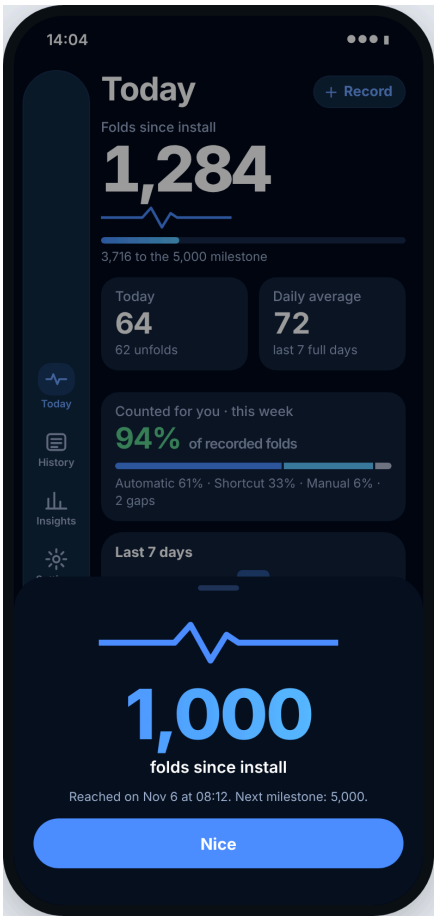
Background counting setup + self-test



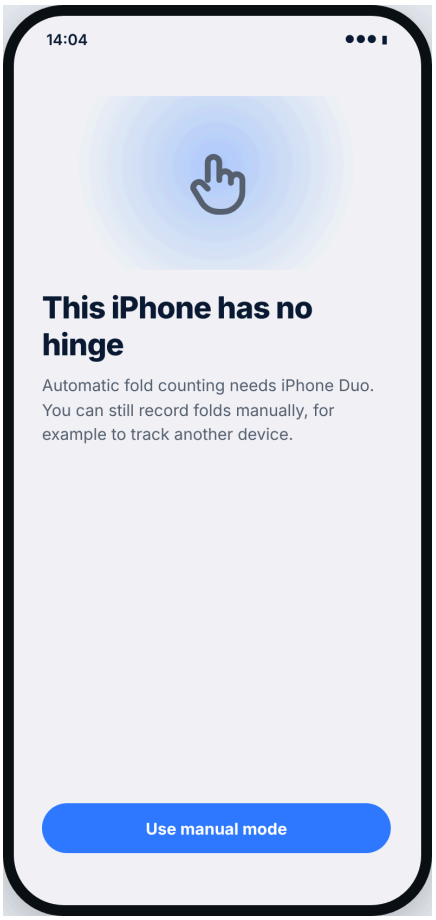
Settings (lean v1)



Export with privacy note

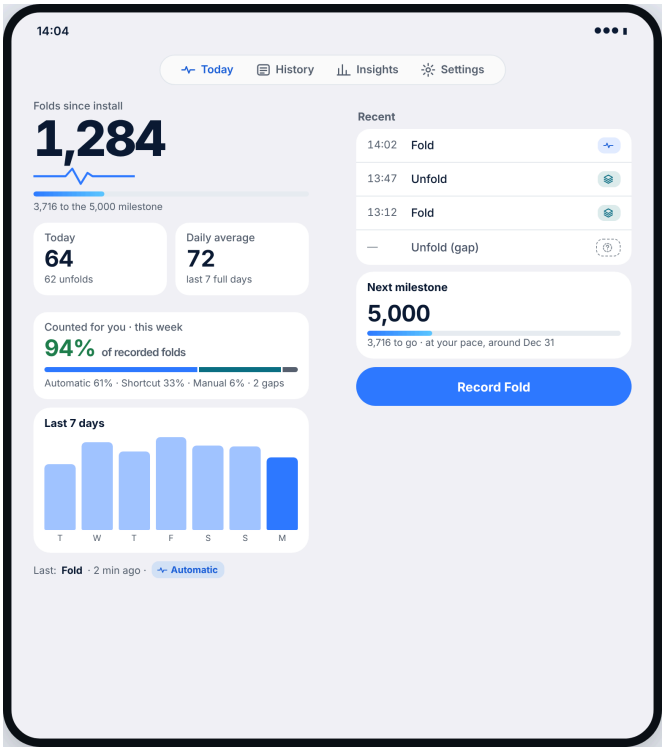


Milestone (once)

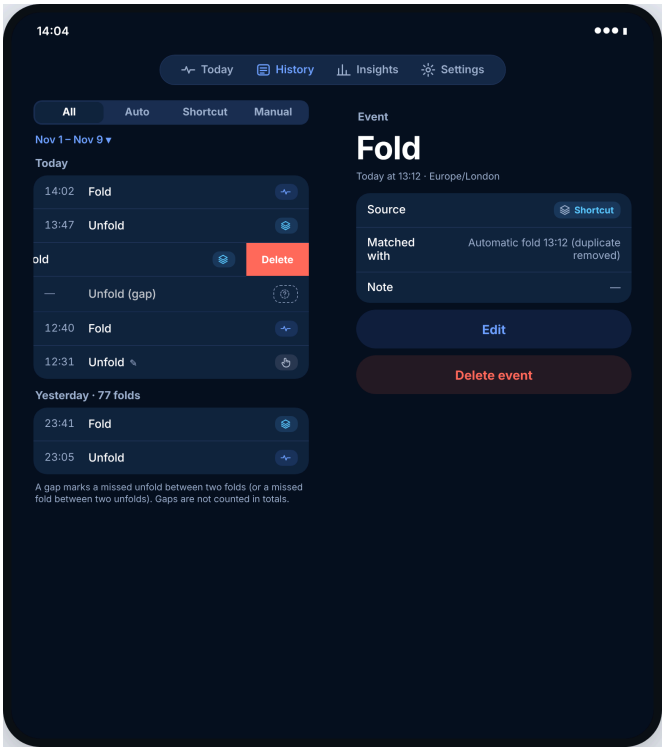


Non-hinge iPhone · manual mode

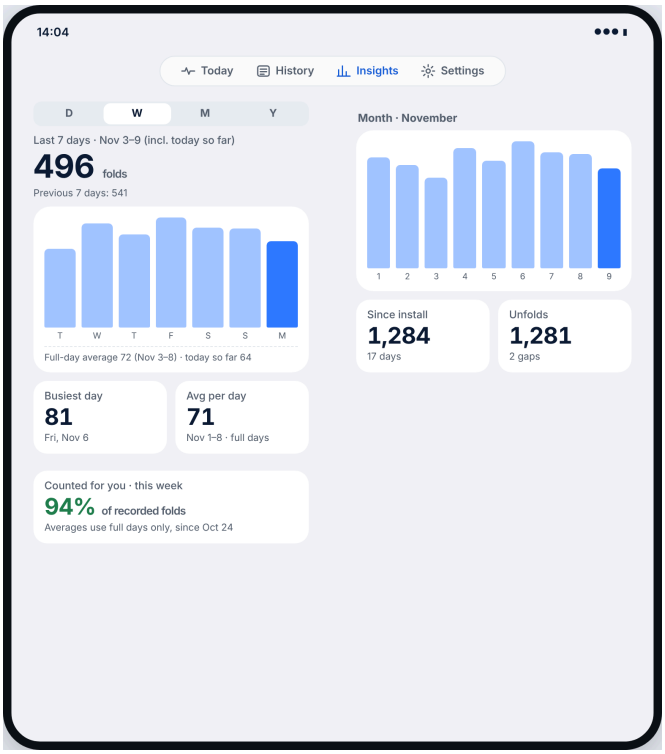
Inner display (7.6")



Inner · Today (ArrangementView split)



Inner · History list + detail



Inner · Insights

Screen specifications

0. Navigation structure

FoldPulse uses a system `TabView` with four tabs and `.tabViewStyle(.sidebarAdaptable)`:

Tab	Purpose	Outer display (compact)	Inner display (regular)
Today	At-a-glance dashboard + record	1 column; the system shows bars vertically on the side	`ArrangementView` split: stats · recent + record
History	Every event, filter, edit, delete	List → detail push	List + detail side by side
Insights	Trends and averages	1 column	2 columns (period · month)
Settings	Counting, appearance, data, about	List	List + detail

Modal surfaces are **sheets**: Record, Edit event, Export, Milestone, Onboarding (full-screen cover on first launch).

General rules:

- The system draws all bars (**Liquid Glass**). No custom tab bars.
- When **partially folded**, content and controls avoid the fold (active `.divisions` reserved region, handled by `ArrangementView`). When fully open there is no division.
- Toolbar items always have **an icon and a title**: on the outer display Apple presents bars vertically, and title-only items aren't shown there. Prominent items use `topBarPinnedTrailing`.
- Every screen supports light and dark mode, Dynamic Type to AX5, and VoiceOver (see DESIGN_SYSTEM §9).

- No screen shows prices, upgrades or ads (D-5, paid up front).

1. Onboarding (3 pages) — `o-onboard-1…3`

Entry	First launch only (`hasCompletedOnboarding == false`)
Page 1	"Every Fold. Every Beat." + animated pulse line + one-sentence value
Page 2	**How counting works**: Automatic (while open) · Shortcut (optional, background) · Manual. "Every number shows where it came from."
Page 3	**Private by design**: data stays on this iPhone, no account/cloud/tracking; export any time
CTA	"Continue" → "Start counting" opens **Today**. Background setup is **not** forced (decision C-2r: it's optional)
Notes	No permission prompts during onboarding. Notification permission is asked only when the user enables milestone notifications
Acceptance	US-01, US-02 AC6 (honest description), PRD §7 privacy

2. Today (dashboard) — `o-today-light/dark` , `i-today`

Element	Content	Behaviour
Title	"Today" (large title)	
`HeroCounter`	**Folds since install**, e.g. 1,284. When background counting is not set up: **"Folds counted since install"**, so a partial count isn't read as complete	`.numericText()` roll on change; pulse beats once per fold
`MilestoneBar`	Progress to the next milestone + "3,716 to the 5,000 milestone"	Uses the milestone list 100 / 500 / 1k / 5k / 10k / 25k / 50k / 100k
Tiles	**Today** (folds + unfolds) · **Daily average** (last 7 complete days)	Today resets at local midnight
`CoverageTile`	"Counted for you · 94% of recorded folds" + stacked bar by source + gap count	Tap → Coverage detail. No "Good" verdict: missed folds can't be in the denominator, so the number describes recorded folds only
Week chart	Last 7 days, today highlighted	VoiceOver: Audio Graph
Last event	"Last: Fold · 2 min ago · [Automatic]"	Relative time updates every minute
Inner display extras	Recent events list, next-milestone card with pace estimate, **Record Fold** button	
Outer display	Record via the toolbar **"+ Record"** item (icon + title, `topBarPinnedTrailing`) → Record sheet	
Hinge observation	While this screen (or any FoldPulse screen) is visible, automatic counting is active	Implementation: one device-level observer (feasibility §5)

States

State	Display
First run / no data (`o-today-empty`)	Hero "0"; "Ready when you are" card; optional **Count in the background** card; Record Fold button

State	Display
Background counting not set up (`o-today-foreground` , the likely default)	Hero label "Folds counted since install"; the coverage tile is replaced by the card **"Counting only while FoldPulse is open"** explaining what's missed + "Set up background counting"
Gaps exist	Coverage tile footnote "2 gaps this week" → History filtered to gaps
Data loading (cold start)	Cached snapshot shown immediately, then refreshed (no spinner)

3. Record sheet — `o-track`

Entry	"+ Record" toolbar item (outer), Record Fold button (inner)
Presentation	Sheet with `.presentationDetents([.medium, .large])` , a **Close** button (`cancellationAction`) and title "Record"
Live status	"Hinge right now: Fully open / Partially open / Closed" (from `onHingeChange`)
Actions	**Record Fold** (primary, 56 pt), **Record Unfold** (secondary)
Feedback	Toast "Fold recorded · Manual" with **Undo** (5 s; persistent while assistive technologies run, see DESIGN_SYSTEM §8) + `.increase` haptic + VoiceOver announcement
Guidance	"Use manual recording only for folds FoldPulse missed. Automatic counts already include folds made while the app is open." This prevents double counting when the phone is folded with the sheet open, which is also handled by de-duplication
Acceptance	US-03 AC1–AC2

4. History — `o-history` , `i-history`

Element	Behaviour
Source filter	Segmented `Picker` : All · Auto · Shortcut · Manual (+ Gaps when any exist)
Date range	Toolbar `Menu` "Nov 1 – Nov 9" → presets + custom `DatePicker`
Sections	Grouped by local day: "Today", "Yesterday · 77 folds", then dates
`EventRow`	Time · Fold/Unfold · source badge (icon-only on the outer display) · ✎ if it has a note or was edited
Gap row	Dashed badge, "Unfold (gap)", no time ("—"). It always sits **between two events of the opposite type** (e.g. between two folds); explanation in the footer
Swipe	`List` + `.swipeActions` → **Delete** (destructive); confirmation for automatic and shortcut events
Tap	Detail (inner: right column; outer: push). Shows source, time zone, de-duplication info ("Matched with automatic fold 13:12"), note, Edit, Delete
Performance	Lazy list; tested to 50,000 events (US-06)

5. Edit event sheet — `o-edit`

Fields: Type (Fold/Unfold), Time (date picker, keeps the original time zone), Source (read-only + "edited"), Note (≤ 500 characters). Buttons: Cancel / Save / Delete event. After saving, reconciliation re-runs (feasibility §6). Acceptance: US-04.

6. Insights — `o-insights`, `i-insights`

Element	Behaviour
Period	Segmented control **D / W / M / Y**
Headline	Total folds in the period ("incl. today so far" when it contains today) + the previous period's total
Chart	Bars per hour (D), day (W, M) or month (Y); average rule line
Tiles	Busiest day · Average per day (full days only). Inner display adds: Since install · Unfolds + gaps
Coverage	"Counted for you" repeated, with the note "Averages use full days only, since "
Deferred to v1.1 (D-6)	Time-of-day heat strip, session durations
Acceptance	US-07 AC1–AC2

7. Coverage detail — `o-coverage`

Hero % for this week ("of N recorded folds"); a row per source (count and %); gaps (not counted); a background-counting status card ("Working · last run 13:47 · self-test passed ", or "Not set up" → Setup, or "No automation run in 24 h" → troubleshooting). Includes a plain-language definition: *the share of recorded folds captured without tapping; folds missed entirely can't be included; gaps show where something was missed.* Acceptance: US-12.

8. Background counting setup — `o-setup`

Entry	Today card, Coverage detail, Settings → Background counting
Content	Why (one sentence, "Optional") · four **plain numbered** steps (no checkmarks: FoldPulse cannot see what the user does in Shortcuts) · **Open Shortcuts** · **Self-test** (the only status shown) · "To stop, delete the two automations in Shortcuts"
Labels to verify	The trigger name ("Opened or Closed"), the "Run Immediately" option and the setup time come from press reports. **Verify them on a device in Phase 5 (S-2)** before shipping the copy
Self-test	The user closes and opens the phone. FoldPulse checks for a `shortcut` fold **and** a `shortcut` unfold within 60 s → "Working ✓"; otherwise it names the failing step
Recovery	"**Can't find the trigger?** It needs iOS 27.2 or later (Settings → General → About). FoldPulse still counts automatically while it's open." (Learned in the Phase 1b spike)
Version gate	If the device runs iOS < 27.2, the screen shows the update notice instead of the steps
Honesty	Never says background counting is on until the self-test passes
Acceptance	US-02 AC1–AC6

9. Settings — `o-settings`

Group	Rows
Counting	Background counting (status: "Not set up" / "Self-test passed" / "Not seen recently" → Setup) · Milestone notifications (off). *"Count gaps in totals" removed: deferred to v1.1 (D-6); gaps are always excluded (D-2)*
Appearance	Theme: System / Light / Dark
Your data	Export data · **Reset all data** (destructive, two-step confirmation, type "RESET"). *Import deferred to v1.1 (D-6)*

Group	Rows
About	Privacy ("Data stays on this iPhone" → privacy statement) · Version · Acknowledgements

10. Export sheet — `o-export`

Format CSV / JSON · range (all time / custom) · privacy card ("never uploads… fold times can reveal daily routines") · **Export** → system share sheet. CSV cells are escaped against formula injection (PRIVACY_REVIEW P-4). *Import is v1.1 (D-6); its flow is kept in USER_FLOWS F4 for later.*

11. Milestone celebration — `o-milestone`

Shown **once** per milestone, the next time the app is opened after crossing it: pulse line draws in, the milestone number in the gradient, date reached, next milestone, one button "Nice". It is never re-shown, even if deletions drop the total (feasibility §6). No confetti, streaks or badges.

12. Manual-only mode — `o-manual-only`

On devices where no hinge reading has ever been received after 10 s on screen (and the model has no hinge), FoldPulse explains: "This iPhone has no hinge. Automatic counting needs iPhone Duo. You can still record folds manually." Widgets and analytics work the same. Coverage is hidden. (D-4; App Review 2.4.1.)

13. Error and edge states (all screens)

Situation	Behaviour
Storage write fails	Non-blocking banner "Couldn't save the last fold. Retrying." The event is kept in the pending queue
Clock or time-zone change	Days re-bucket per the stored `localDate` rules; no user-facing message
Import file invalid	Sheet with a specific reason; nothing is written
Data reset	Returns to the first-run Today state; widgets reload

7 Widgets

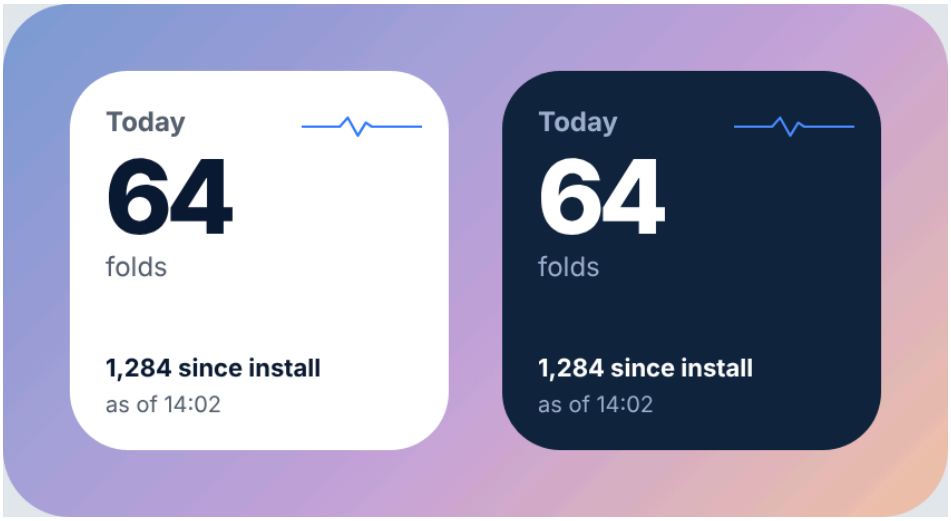


Figure 5 — Home Screen small widget, light and dark

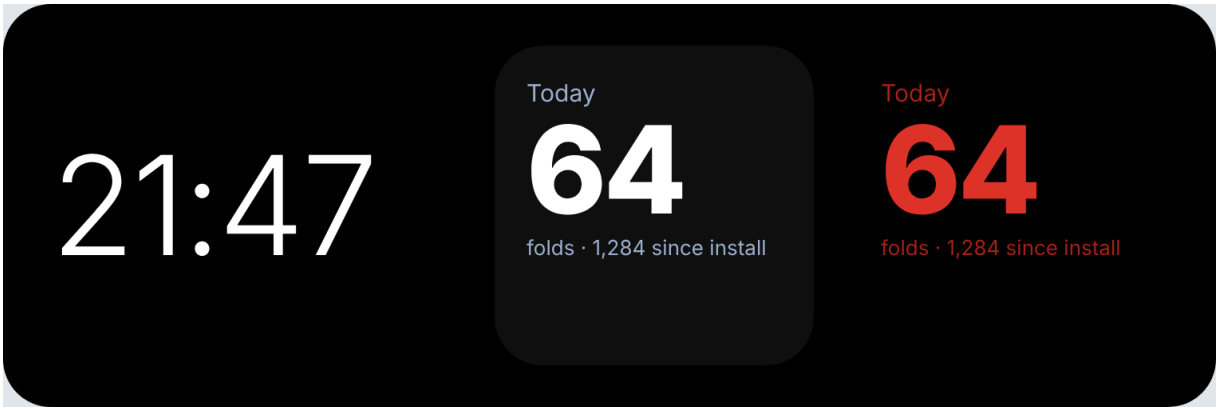


Figure 6 — StandBy (system-scaled small widget; right: night mode)

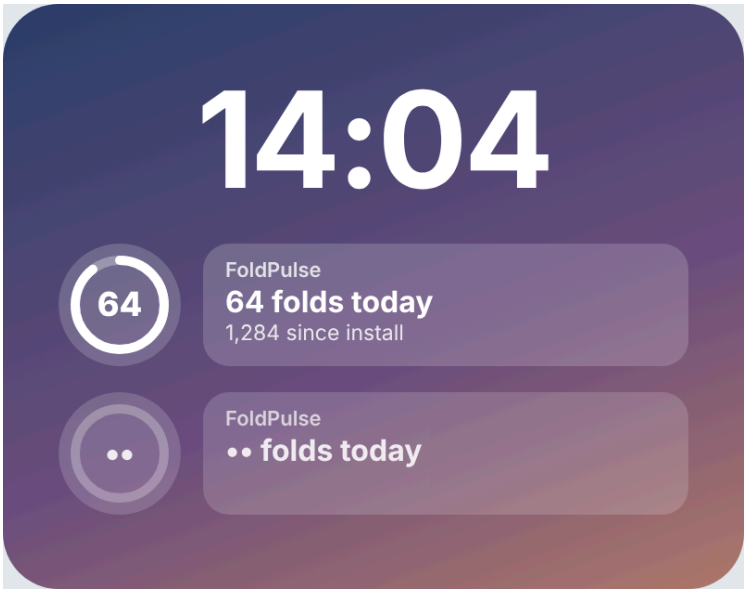


Figure 7 — Lock Screen circular and rectangular; bottom row: privacy-redacted (approximate)

1. Lean v1 widget set (decision D-6)

Widget	Family	Content	Mockup
Today (Home Screen)	`systemSmall`	"Today" · today's folds (hero) · "folds" · "1,284 since install" (never "total") · freshness · pulse mark	`w-home` (light & dark)
StandBy	`systemSmall` presented by the system in StandBy	Same widget. The system scales it and applies the night-mode red tint	`w-standby`
Today ring (Lock Screen)	`accessoryCircular`	Today's folds inside a gauge of today ÷ 7-day daily average (capped at 100%)	`w-lock`
Today + total (Lock Screen)	`accessoryRectangular`	"FoldPulse" · "64 folds today" · "1,284 since install"	`w-lock`

Deferred to v1.1: `systemMedium`, `systemLarge`, `accessoryInline`, and the interactive "+ Fold" button. The button is also inactive while the phone is locked (an Apple restriction) and runs in the widget process (risk R-18).

2. Data and freshness

- The widget reads **`WidgetSnapshot.json`** from the App Group (file protection **`completeUntilFirstUserAuthentication`**, so it stays readable on the Lock Screen and writable by background intents while locked): `todayFolds`, `todayUnfolds`, `totalSinceInstall`, `dailyAverage7d`, `nextMilestone`, `localDate`, `generatedAt`. It never opens the database.
- **Freshness label:** "as of 14:02" whenever `now − generatedAt > 15 min`. Otherwise no label (Small) or "Updated just now" (in the app only).
- **Midnight rule:** the timeline contains a second entry at the next local midnight. From that moment "Today" shows **0** and the label reads "as of 23:58 yesterday" for the total. No reload is needed (Apple: predictable timeline events).
- **Reload policy (Agent 11):** reload immediately when the app is in the foreground (unbudgeted). From background intents, reload at most once every 5 minutes (coalesced). The budget is roughly 40–70 reloads a day.
- **Empty state:** "0 · Start folding" before any data. Tapping the widget opens Today.

3. Rendering modes

Mode	Treatment
Full colour (Home)	`containerBackground` = `surface` token; numerals `text`; pulse mark `accent`
Accented / tinted Home Screen	Numerals and pulse marked `.widgetAccentable()`; background removed by the system
Vibrant (Lock Screen, StandBy night)	Monochrome; hierarchy via opacity (primary 100%, secondary 70%)
Clear / Liquid Glass Home Screen styles	No custom background images; rely on system materials

Checked with `@Environment(\.widgetRenderingMode)`.

4. Privacy

- Lock Screen counts **and the gauge value** use `.privacySensitive()`. The system draws the redaction, so the "••" in `w-lock` (bottom row) is an approximation, to be checked on a device.
- Widgets show counts only, never timestamps of individual folds.

5. Accessibility

- VoiceOver label: "FoldPulse. 64 folds today. 1,284 since install. Updated 14:02."
- Numerals use Dynamic Type within widget limits. In the circular family the number shrinks to fit (`minimumScaleFactor(0.6)`) rather than truncating.
- The gauge also announces its value: "64 of your daily average of 72".

6. StandBy notes

StandBy shows Home Screen small widgets when an iPhone is charging, locked and on its side. **On iPhone Duo, which pose (closed or open) enters StandBy is unverified** (spike observation O-4 was skipped). FoldPulse adds nothing StandBy-specific beyond a widget that reads well at a distance: one hero number, two short labels. **It does not claim to replace or customise StandBy.**

8 Accessibility review

Area	Commitment	How it's verified
Contrast	All 36 text/background pairs ≥ 4.5:1; raw accent used only for graphics and large text (≥ 3:1)	**Measured**: `design/contrast_check.py` (table below)
Dynamic Type	To AX5; tiles stack; hero uses `ViewThatFits` + a scaled-size ceiling	Phase 4 UI tests at AX5 on both displays
VoiceOver	Sentence labels; Audio Graphs for charts; icon-only badges labelled; pulse hidden	Phase 5 VoiceOver walkthrough
Timing	Undo persists while assistive tech is running; History offers Undo any time	Phase 5
Motion	Reduce Motion → crossfades; no auto-playing loops	Phase 5
Colour independence	Every source and status carries a symbol and a word	Design review (done)
Targets	≥ 44 pt; Record buttons 56 pt	Phase 4
Privacy on the Lock Screen	`.privacySensitive()` on counts and the gauge	Phase 5 on device

Measured contrast (WCAG 2.x)

Mode	Foreground	Background	Ratio	Required	Result
light	`text` #0B1B33	`bg` #F2F4F8	15.65:1	4.5:1	Pass
light	`text` #0B1B33	`surface` #FFFFFF	17.23:1	4.5:1	Pass
light	`text2` #55606E	`bg` #F2F4F8	5.81:1	4.5:1	Pass
light	`text2` #55606E	`surface` #FFFFFF	6.39:1	4.5:1	Pass
light	`accentText` #1D5FD8	`bg` #F2F4F8	5.18:1	4.5:1	Pass
light	`accentText` #1D5FD8	`surface` #FFFFFF	5.71:1	4.5:1	Pass
light	`shortcut` #0B6F85	`bg` #F2F4F8	5.26:1	4.5:1	Pass
light	`shortcut` #0B6F85	`surface` #FFFFFF	5.79:1	4.5:1	Pass
light	`manual` #55606E	`bg` #F2F4F8	5.81:1	4.5:1	Pass
light	`manual` #55606E	`surface` #FFFFFF	6.39:1	4.5:1	Pass
light	`positive` #1E7F4F	`bg` #F2F4F8	4.54:1	4.5:1	Pass
light	`positive` #1E7F4F	`surface` #FFFFFF	5.00:1	4.5:1	Pass
light	`warning` #A15C00	`bg` #F2F4F8	4.71:1	4.5:1	Pass
light	`warning` #A15C00	`surface` #FFFFFF	5.19:1	4.5:1	Pass
light	`destructive` #C2362B	`bg` #F2F4F8	4.95:1	4.5:1	Pass
light	`destructive` #C2362B	`surface` #FFFFFF	5.45:1	4.5:1	Pass

Mode	Foreground	Background	Ratio	Required	Result
light	`accent` #2F7BFF	`bg` #F2F4F8	3.53:1	3.0:1	Pass
light	`accent` #2F7BFF	`surface` #FFFFFF	3.89:1	3.0:1	Pass
dark	`text` #FFFFFF	`bg` #070F1F	19.14:1	4.5:1	Pass
dark	`text` #FFFFFF	`surface` #12233F	15.70:1	4.5:1	Pass
dark	`text2` #9FB1CF	`bg` #070F1F	8.81:1	4.5:1	Pass
dark	`text2` #9FB1CF	`surface` #12233F	7.22:1	4.5:1	Pass
dark	`accentText` #6FA2FF	`bg` #070F1F	7.55:1	4.5:1	Pass
dark	`accentText` #6FA2FF	`surface` #12233F	6.19:1	4.5:1	Pass
dark	`shortcut` #5CC8FF	`bg` #070F1F	10.17:1	4.5:1	Pass
dark	`shortcut` #5CC8FF	`surface` #12233F	8.34:1	4.5:1	Pass
dark	`manual` #B4C0D3	`bg` #070F1F	10.41:1	4.5:1	Pass
dark	`manual` #B4C0D3	`surface` #12233F	8.54:1	4.5:1	Pass
dark	`positive` #5BD08E	`bg` #070F1F	9.90:1	4.5:1	Pass
dark	`positive` #5BD08E	`surface` #12233F	8.12:1	4.5:1	Pass
dark	`warning` #FFB547	`bg` #070F1F	10.89:1	4.5:1	Pass
dark	`warning` #FFB547	`surface` #12233F	8.93:1	4.5:1	Pass
dark	`destructive` #FF6B5E	`bg` #070F1F	6.85:1	4.5:1	Pass
dark	`destructive` #FF6B5E	`surface` #12233F	5.62:1	4.5:1	Pass
dark	`accent` #4C8DFF	`bg` #070F1F	5.98:1	3.0:1	Pass
dark	`accent` #4C8DFF	`surface` #12233F	4.90:1	3.0:1	Pass

Failures: 0

9 Agent reviews, issues & disagreements

UI Developer (Agent 5) and Compliance & Accessibility (Agent 10)

Finding	Severity	Resolution
Setup screen ticked steps FoldPulse cannot observe; "switch it off any time" implied an in-app switch	High	Plain numbered steps; self-test is the only status; "delete the automations to stop"
Outer display shows bars vertically: toolbar "+" was missing; Record sheet had no Close	High	"+ Record" item (icon + title, pinned trailing); Close + detents
The fold is reserved only when partially open	Medium	`ArrangementView` split; rule corrected
Gap toggle and Import contradicted D-6	Medium	Removed from v1 (tagged v1.1)
"Captured automatically" clashed with the "Automatic" source	Medium	Renamed "Counted for you"
Custom chips, toast and buttons	Medium	Segmented Picker, Menu + DatePicker, List swipeActions, glassProminent buttons
Widget file protection while locked	Medium	`completeUntilFirstUserAuthentication` stated
Icon: Dark ≈ Default; reads as a book at 40 px; pulse + "Pulse" may look like a health app	Medium	Redesigned glyph, true dark and tinted; Utilities category, no health keywords (R-27)
5 s Undo is a timing barrier	Medium	Persistent Undo with assistive tech + announcement
"••" redaction is system-drawn	Medium	Gauge also private; mockup marked approximate
AX5 hero cannot wrap; "1,284 total" wording; untokened beige; unlabelled icon badges	Low	All fixed

Founder Representative (Agent 2)

Finding	Severity	Resolution
A paid app whose core value (background counting) is unproven	Critical	Recorded as **R-28 (High, Founder decision)**: run S-2/S-3 on iOS 27.2 before setting a price. Design made honest for the default state
Mockups showed only the happy path	High	Added `o-today-foreground` with "Folds counted since install" and the counting-only-while-open card
Gap toggle and Import over-built vs lean v1	Medium	Removed (v1.1)
No App Store screenshot concept for a paid app	Medium	Explicitly deferred to Phase 6 (I-25)
Three coverage levels, two tokens	Low	Levels mapped to positive / text2 / warning

Independent Final Reviewer (Agent 12)

Finding	Severity	Resolution
Coverage overclaims (missed folds not in the denominator)	High	"% of recorded folds"; no verdict word; PRD wording fixed
Setup mockup claimed impossible knowledge; trigger labels unverified	High	Fixed; labels marked "verify in S-2"
Sample data inconsistent	Medium	One seed dataset; weekday error (Nov 9 is a Monday) fixed
PRD and MVP out of date (3 s dedupe, deferred widgets, monetization)	Medium	Aligned with D-5, D-6, C-2r
Insights mockup vs spec mismatch; 9:41 status bars	Medium/Low	Aligned; 14:04 throughout
Recommendation: approve with changes	—	Changes applied in this version

Disagreements recorded

1. **Inner tab bar over the fold when partially open (I-24).** UI Developer: acceptable, because the system draws and positions it. Compliance: validate in the partially open pose first. **Resolution:** check in Xcode Device Hub at the start of Phase 3.
2. **Pricing before verification (R-28).** Founder Rep: don't set a price until S-2/S-3 pass. Recorded as a Founder decision; the design team can't resolve it.

10 Decisions requested & next steps

Decisions requested

#	Decision	Team recommendation
A-2	**Approve Phase 2** (design system, icon, screens, flows, widgets)	Approve
R-28	Before setting a price: update your iPhone Duo to iOS 27.2 and run the 10-minute automation check (SPIKE_PLAN Parts 2–3, runs C and D)	Strongly recommended, not blocking Phase 3
I-24	Confirm that the inner tab bar is checked in the simulator in Phase 3	Yes

What Phase 3 will do (after approval)

Architecture & project foundation:

- Create the Xcode project (app, widget extension, App Group), using your Mac via the same one-line script approach.
- Implement the data model and SwiftData persistence with file protection.
- Implement the design tokens and core components in SwiftUI.
- Set up unit and UI tests.
- Verify the build.
- Write ARCHITECTURE.md, DATABASE_DESIGN.md, DECISIONS updates and `phase-03-architecture.pdf`.

Because this cloud workspace can't run Xcode, each build will be verified on your Mac (or a macOS CI runner if you'd like one set up).

11 References

- Apple — Preparing your app for iPhone Duo: https://developer.apple.com/documentation/technologyoverviews/preparing-your-app-for-iphone-duo
- Apple — Human Interface Guidelines: https://developer.apple.com/design/human-interface-guidelines/
- Apple — Keeping a widget up to date: https://developer.apple.com/documentation/widgetkit/keeping-a-widget-up-to-date
- Apple — WidgetFamily: https://developer.apple.com/documentation/widgetkit/widgetfamily
- Apple — Adding interactivity to widgets: https://developer.apple.com/documentation/widgetkit/adding-interactivity-to-widgets-and-live-activities
- Apple — SwiftUI onHingeChange(isEnabled:_:): https://developer.apple.com/documentation/swiftui/view/onhingechange(isenabled:_:)
- Apple — App Store Review Guidelines: https://developer.apple.com/app-store/review/guidelines/
- W3C — WCAG 2.2 (contrast 1.4.3, timing 2.2.1): https://www.w3.org/TR/WCAG22/
- Project files: `docs/DESIGN_SYSTEM.md`, `docs/SCREEN_SPECIFICATIONS.md`, `docs/USER_FLOWS.md`, `docs/WIDGET_DESIGN.md`, `design/` (mockups, flows, icon, contrast check)